INCH POUND

MIL-T-28919A(MC)

20 April 1994

SUPERSEDING
MIL-T-28919(MC)
23 April 1969

# MILITARY SPECIFICATION

## TROUSERS, MEN'S: SERVICE, SUMMER AND WINTER (OFFICER'S)

This specification is approved for use by U. S. Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers requirements for men's polyester/wool trousers.

1.2 Classification. The trousers shall be of one type, in the following styles and sizes as specified (see 6.2):

Polyester/Wool Gabardine, Green Shade 2212

Styles

Style A - With watch pocket
Style B - Without watch pocket

Schedule of Sizes

| Waist | 26 | 28 | 30 | 32 | 34 | 36 | 38 | 40 | 42 | 44 | 46 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Inseam Lengths | X-Short, Short, Regular, Long and X-Long | | | | | | | | | | |

1.3 Individual orders. When trousers are custom tailored to an individual order, this specification shall govern, except the detail requirements of Table I of Section 3 may be modified to the extent necessary for good custom tailoring.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commander (PSC), Marine Corps Systems Command, Quantico, VA 22134-5010 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8405

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## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| A-A-50186 | - Cloth, Buckram, Woven and Nonwoven |
| A-A-50198 | - Thread, Gimp, Cotton, Buttonhole |
| A-A-50199 | - Thread, Polyester Core: Cotton; or Polyester-Covered |
| A-A-52094 | - Thread, Cotton |
| A-A-52106 | - Cloth, Twill or Plain Weave, Polyester or Polyester Blend |
| V-B-871 | - Button, Sewing Hole, and Button, Staple (Plastic) |
| V-F-106 | - Fasteners, Slide, Interlocking |
| V-T-295 | - Thread, Nylon |
| UU-P-268 | - Paper, Kraft, Wrapping |
| DDD-L-20 | - Label; For Clothing, Equipage, and Tentage, (General Use) |

MILITARY

| | |
|---|---|
| MIL-C-10176 | - Cloth, Gabardine: Wool, Polyester and Wool |
| MIL-B-17757 | - Box, Shipping, Fiberboard (Modular Sizes) |
| MIL-C-43920 | - Cloth, Interlining, Cotton or Synthetic and Nylon |

STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-751 | - Stitches, Seams and Stitchings |

MILITARY

| | |
|---|---|
| MIL-STD-129 | - Marking for Shipment and Storage |
| MIL-STD-1488 | - Provisions for Evaluating Quality of Trousers |

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government publications. The following other Government publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.



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Department of Defense (DoD) Standard Color Card of Official Standardized Shades for Sewing Threads 1986

Department of Defense (DoD) Standard Shades for Buttons 1966

(Application for copies should be addressed to the Color Association of the United States Inc., 200 Madison Avenue, New York, New York 10016.)

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF TESTING AND MATERIALS

ASTM D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents may also be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 Guide sample. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.4 and 6.4).

3.3 Materials

3.3.1 Basic material. The basic material for the trousers shall be cloth, 10.6 ounce polyester/wool gabardine green, shade 2212 conforming to type II, class 12, of MIL-C-10176.



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3.3.2 Lining, interlining and pocketing.

3.3.2.1 Lining and interlining. The material for lining the waistband, right fly, pocket flaps, crotch, left fly interlining, and pocket stay shall be black polyester, polyester and cotton, or polyester and rayon cloth conforming to class 1 of A-A-52106.

3.3.2.2 Pocketing. The pocketing material shall be black poly ester, polyester and cotton or polyester and rayon cloth conforming to class 1 of A-A-52106.

3.3.3 Buckram. The material for interlining waistband shall be bleached or unbleached cloth, cotton, buckram conforming to type II, class 1 of A-A-50186.

3.3.4 Waistband stabilizer. The material for the waistband stabilizer shall be cloth, interlining, cotton or synthetic and nylon conforming to type 1 of MIL-C-43920.

3.3.5 Slide fastener. The slide fastener for each class of trousers shall be type I, style 1, cam lock, size LS, short tab pull, brass chain, conforming to V-F-106, except the salt spray test shall not apply. A staple bottom stop is required. All components of the slide fastener shall be brass and from same supplier (see 4.5.1). The tape shall be cotton or synthetic or a combination.

3.3.5.1 Color. The color of the slide fastener tape shall be Green, Sh AF, Cable No. 66531.

3.3.5.2 Colorfastness. The dyed slide fastener tape shall show colorfastness to wet dry-cleaning equal to or better than the standard sample when tested as specified in V-F-106. When no standard sample is available, the tape shall show "good" colorfastness when tested as specified in V-F-106.

3.3.6 Hook and eye. The hook and eye shall be North and Judd (Hook Flex H-49, K-50, H-89 and E-90); Universal Button Co. (Talon #3, #7 and #8 hook with #8 eye or #85 eye); Akim Co. Or Patwin, Inc., Junior Series, type JB; Stimpson (P-536, D-4086, D-4457, D-4288); Rau (OA50, OX 1256, 080415), Patwin, Inc. (Slim-Slak with MB eye, and Slik-Slak) or equal. The finish shall be nickel plate. Prior to the use of an "or equal" item, the supplier shall submit the item with supporting data to the contracting officer for approval (see 4.5.1).

3.3.7 Thread. Cotton thread shall be used conforming to A-A-52094 as follows:



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| Operation | Type | Ticket No. | Ply |
|---|---|---|---|
| Outside stitching, bartacks, buttonholes and ornamental strip. | III | 0 | 3 |
| Seaming and Stitching | I | 30, 50 | 3 |
| | I | 70 | 2 |
| Automatic machine pocket construction | III | A | 3 |
| Inside Stitching | I | 50 | 3 |
| | | 70 | 2 |
| Button Sewing | II | 16, 20 | 4 |

As an alternate to the cotton thread, a polyester, cotton-covered thread conforming to A-A-50199, ticket No. 50, 3 ply or ticket No. 70, 2 ply may be used for all sewing operations.

3.3.7.1 Automatic button sewing. If an automatic machine which sews and shanks (wraps) the buttons in one operation is used, the thread shall be waxed nylon monocord special No. 5, having a minimum breaking strength of 16 pounds (7.3 kg) and minimum length per pound of 2150 yards (1965 m). This thread shall be tested for these requirements and for colorfastness (see 3.3.7.3) in accordance with test methods in V-T-295.

3.3.7.2 Color. The color of the thread for the trousers shall be Olive Drab, T-1, Cable No. 66023.

Note: See DoD Standard Color Card of Official Standardized Shades for Sewing Threads 1968.

3.3.7.3 Colorfastness. The colored thread shall show fastness to light and wet dry-cleaning equal to or better than the standard sample. When no standard sample is available, the colored thread shall show "good" fastness to light and wet dry cleaning.

3.3.8 Gimp. The gimp for reinforcing buttonholes shall be cotton, No. 8, conforming to type I or II of A-A-50198.

3.3.8.1 Color. The color requirements for the gimp for the trousers shall be the same as specified in 3.3.7.2 for the thread.

3.3.8.2 Colorfastness. The colored gimp shall show fastness to wet dry-cleaning equal to or better than the standard sample. When no standard sample is available, the colored gimp shall show "good" fastness to wet dry-cleaning.

3.3.9 Buttons. The buttons for the hip pocket and fly-tab closure shall conform to type II, class D, style 4, 22-line, glossy finish, of V-B-871. The color of the buttons for the trousers shall be Green, Shade AF, Cable No. 62017 (see DoD Standard Shades for Buttons, 1966).



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3.3.10 Labels. Each pair of trousers shall have a combination identification-size-instruction label. The combination label shall conform to type VI, class 14 of DDD-L-20. The label shall conform to the colorfastness and writing test requirements of DDD-L-20. The content (inscription) and format of the combination label shall be as follows:

NOMENCLATURE:
Trousers, Man's
Fiber Content: 55% Polyester/45% Wool (Example)
Size: 38R (Example)
Name of Contractor:
DRY CLEAN ONLY

3.4 Design. The trousers shall be U. S. Marine Corps design as shown on Figure 1 having a two-piece, lined waistband with hook and eye closure and outlet at seat seam; french type fly with slide fastener; 7 belt loops for sizes up to and including size 37 and 8 belt loops for sizes 38 and up; straight side pockets; hip pockets with pocket flaps to button; outlet from crotch to knee at inseam and unfinished leg bottom.

3.5 Patterns. Standard patterns will be furnished to the contractor by the Government. The standard patterns shall not be altered in any way and shall be used only as a guide for cutting the contractors working pattern. The contractor's working patterns provide an allowance of 1/4 inch for pocket flaps, and a 3/8 inch seam allowance for all other seams except as otherwise stated operation.

3.5.1 Pattern parts. The component parts of the trousers shall be cut from the materials specified herein in accordance with the number of pattern parts indicated below, except belt loops and waist band lining/interlining. (See Table I, operation 1, for cutting requirements for these components).

| Item | Pattern Nomenclature<br>Material: Basic | No. of cut parts |
|---|---|---|
| 1 | Front | 2 |
| 2 | Back | 2 |
| 3 | Right fly | 1 |
| 4 | Left fly | 1 |
| 5 | Side pocket wide facing (bearer) | 2 |
| 6 | Side pocket narrow facing | 2 |
| 7 | Hip pocket bearer | 4 |
| 8 | Hip pocket flap | 2 |
| 9 | Waistband | 2 |



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| Item | Pattern Nomenclature | No. of cut parts |
|---|---|---|
| | Material: Lining/Interlining | |
| 10 | Right fly lining | 1 |
| 11 | Crotch lining | 2 |
| 12 | Side pocket stay | 2 |
| 13 | Left fly interlining | 1 |
| 14 | Hip pocket flap lining | 2 |
| | Material: Pocketing | |
| 15 | Side pocket | 2 |
| 16 | Hip pocket | 2 |

3.6 Construction. The trousers shall be manufactured in accordance with all operations specified in Table I. The contractor is not required to follow the exact sequence of operations, but all operations specified shall be used to manufacture the trousers. Any additional basting or holding stitching used to facilitate manufacture is permissible provided that the thread is removed or does not show on the finished trousers.

3.6.1 Stitches, seams, and stitchings. Stitch, seam and stitching types specified in Table I shall conform to FED-STD-751. Whenever two or more methods for seam or stitch types are given for the same part of the operation, any one of them may be used. Where stitch type 401 is used, the looper (under thread) shall be on the inside of the trousers. Seam allowances shall be maintained with seams sewn so that no raw edges, runoffs, twists, pleats, puckers or open seams will result. All seams shall start and finish evenly. Thread tensions shall be maintained so that there will be no tight or loose tension. Buttonholes shall be clean cut and the stitching shall be securely caught in the fabric.

3.6.1.1 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table I.

3.6.1.2 Thread breaks and ends of seam. The ends of all seams and stitching, when not caught in other seams and stitching, shall be backtacked not less than 1/4 inch. The ends of a continuous line of stitching (except labels) shall be overlapped not less than 1/2 inch. The ends of the label stitching shall be overlapped not less than three stitches. Thread breaks (all stitch types) shall be secured by stitching back of the break not less than 1/2 inch. Skipped stitches or thread breaks on 401 seaming may be repaired using 301 stitch type. Bartacks shall be not more than 1/16 inch wide.

3.6.2 Buttonholes. Buttonholes shall be the eyelet-end, taper-bar type, well worked over gimp with the end securely tacked. The finished cut length shall be 5/8 inch. The bar at end of the buttonhole shall have a minimum of four tacks, not including the first crossover stitch.



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3.6.3 Pressing. Pressing as required in Table I shall be accomplished with a heated pressing iron or machine.

3.6.4 Marking. The component parts shall be marked to insure a uniform shade and size throughout the trousers. Any method of shade marking may be used except:

(a) Corrosive metal fastening devices.
(b) Sew-on shade tickets.
(c) Adhesive type tickets, which leave traces of paper or adhesive on material after removal of the tickets.

TABLE I MANUFACTURING REQUIREMENTS

| NO | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH/ IN | NDL | BOB/ LPR |
|---|---|---|---|---|---|---|
| 1 | Cutting.<br><br>a. Cut the trousers in strict accordance with the patterns furnished by the Government.<br><br>b. Cut the material in the direction indicated by directional lines on the patterns. Lay fronts and backs so that the directional lines are in the wrap direction.<br><br>c. Cut all component parts for the trousers from the same piece of basic material to insure a uniform shade throughout the trousers. The following parts may be cut from ends but when cut from ends the parts shall harmonize with the shade of trousers: belt loops, pocket bearers and facings.<br><br>d. Cut the pockets on the straight of the material in accordance with patterns. Cutting of pockets in the direction of the filling is not permitted except when it is essential for economical use of the material.<br><br>e. Cut stripping for belt loops of sufficient width to comply with the requirements of operation 11.<br><br>f. Cut the right fly lining in accordance with the directional lines on the pattern. | | | | | |



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TABLE I MANUFACTURING REQUIREMENTS

| NO | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH/ IN | NDL | BOB/ LPR |
|---|---|---|---|---|---|---|
| 1 | Cutting. (cont'd)<br><br>g. Cut left fly interlining, hip pocket flap linings, and stays for side pockets in the warp or filling direction.<br><br>h. Cut crotch lining parts so that, when folded, the fold will be on the bias, and one of the inseam sides is in the warp direction.<br><br>i. Cut the waistband linings and interlinings on the bias and of sufficient width to comply with operation 13.<br><br>j. Cut waistband stabilizer of sufficient length to comply with operation 13.<br><br>k. The waistband lining may be commercial type comprised of the lining, interlining and waistband stabilizer materials specified. | | | | | |
| 2 | Replacement of damaged parts.<br><br>a. At time of cutting, replace any part containing damages such as holes or seriously weakening defects such as smashes, multiple floats and slubs, likely to develop into holes, as classified in MIL-STD-1488.<br><br>b. Replace any part damaged during the manufacturing process by needle chews which may be expected to develop into holes, scissors or knife cuts, tears, holes, mends, burns and large exposed drill holes. | | | | | |



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TABLE I MANUFACTURING REQUIREMENTS

| NO | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH/ IN | NDL | BOB/ LPR |
|---|---|---|---|---|---|---|
| 3 | Shade markings. | | | | | |
| | a. Mark or ticket the component parts to insure a uniform shade and proper assembly throughout the trousers, except those parts specified in operation 1c. which may be cut from end and need not match each other. Tickets shall conform to 3.6.4. | | | | | |
| 4 | Overedge stitch trousers. | | | | | |
| | Finished appearance. The overedge stitching shall be securely caught in the fabric. The raw edge of the overedged material shall be properly trimmed. | | | | | |
| | a. Overedge stitch the inseam of front trousers, also the fly edge of right front. The overedge stitching of the inseam of front shall extend from bottom of trouser to fly notch. The overedge stitching of the outseam of fronts shall extend from bottom notch of side pocket to bottom of trouser leg. | 502 | EFd-1 | 6-10 | 70/2 | 70/2 |
| | | or 503 | EFd-1 | 6-10 | 70/2 | 7 |
| | | or 504 | EFd-1 | 6-10 | 70/2 | 70/2 |
| | b. Overedge stitch inseam and outseam of back part of trousers. The side pocket stay piece with the bottom edge turned up may be attached to the outseam in the overedge operation. | 502 | EFd-1 | 6-10 | 70/2 | 70/2 |
| | | or 503 | EFd-1 | 6-10 | 70/2 | 70/2 |
| | | or 504 | EFd-1 | 6-10 | 70/2 | 70/2 |
| | c. Overedge stitch crotch seam, seat seam and raw edge at the left end of waistband. | 502 | EFd-1 | 6-10 | 70/2 | 70/2 |
| | | or 503 | EFd-1 | 6-10 | 70/2 | 70/2 |
| | | or 504 | EFd-1 | 6-10 | 70/2 | 70/2 |
| | d. The folded crotch lining may be stitched to the foreparts in a separate operation (see oper. 13). | 301 | SSa-1 | 10-14 | 50/3 | 50/3 |



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| NO | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH/ IN | NDL | BOB/ LPR |
|---|---|---|---|---|---|---|
| 4 | Overedge stitch trousers. (cont'd)<br><br>Note: When a knife is used on serging (overedge stitching) machine, the gage shall be set so that the raveled ends are trimmed but the edge of the fabric is not cut off. | | | | | |
| 5 | Set hip pocket bearer.<br><br>Finished appearance. The hip pocket bearer shall be properly positioned, uniformly and securely sewn with stitching of uniform gage and without pleats, puckers or other distortions.<br><br>a. The hip pocket bearers shall each be a single ply of basic material.<br><br>b. Position the bearer on the pocketing material in accordance with notches as indicated by marks on the pattern. | | | | | |
| | c. Fold under the lower raw edge of the hip pocket bearer uniformly 1/4 to 3/8 inch and stitch to the pocket 1/16 to 1/8 inch from the folded edge. When bearer is cut on selvage, the edge need not be turned under. | 301<br>or<br>301 | LSd1-1<br><br>LSdj-1 (selvage) | 10-14<br><br>10-14 | 50/3<br><br>50/3 | 50/3<br><br>50/3 |
| 6 | Set side pocket bearer and facing.<br><br>Finished appearance. The side pocket bearer and facing shall be properly positioned and securely sewn with stitching of uniform gage and without pleats, puckers or other distortion.<br><br>a. The side pocket bearer and facings shall each be a single piece of basic material. | | | | | |



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TABLE I MANUFACTURING REQUIREMENTS

| NO | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH/ IN | NDL | BOB/ LPR |
|---|---|---|---|---|---|---|
| 6 | Set side pocket bearer and facing. (cont'd) | | | | | |
| | b. Position the bearer and facing on the pocket material in accordance with notches as indicated by marks on the pattern. | | | | | |
| | c. Fold under the front raw edge of bearer and facing 1/4 to 3/8 inch and stitch to the pocket 1/18 from the folded edge. Continue stitching across the bottom raw edge. A selvage edge need not be turned. | 301 | LSd-1 and LSj-1 | 10-14 10-14 | 50/3 50/3 | 50/3 50/3 |
| 7 | Make hip pocket flaps. | | | | | |
| | Finished appearance. The hip pocket flaps shall finish smooth and flat, without twists, pleats, or puckers and shall be uniform in shape and size in accordance with patterns. The lining shall not be exposed on the outside. The edges of the flaps shall be well forced out and the edgestitching shall be uniform in width. The button holes shall be correctly positioned, clean cut, and well made, and the stitching securely caught in the basic material. | | | | | |
| | a. Flaps for the hip pockets shall be made of one ply of basic material and one ply of lining material. - or - 2 plies of basic material may be used. | | | | | |
| | b. Stitch the two plies of material together 3/16 to 1/4 inch from the side and bottom edges. Evenly trim the edges for proper turning when necessary. | 301 or 401 | SSe-2 (a) SSe-2 (a) | 10-14 10-14 | 0 0 | 0 0 |



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TABLE I MANUFACTURING REQUIREMENTS

| NO | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH/ IN | NDL | BOB/ LPR |
|---|---|---|---|---|---|---|
| 7 | Make hip pocket flaps. (cont'd) | | | | | |
| | c. Turn the flaps, working out the seamed edges, corners and points completely. Edge stitch the seamed edges of flaps uniformly 1/16 to 3/32 inch from seamed edges, working the lining or under ply back from the finished edge of the flaps. The lining underply shall not be exposed on the outside. | 301 | SSe-2 (b) | 10-14 | 0 | 0 |
| | d. Make a vertical, eyelet end, taper bar type buttonhole cut 5/8 inch long hole through the center of the flap with the edge of the eyelet 1/2 to 5/8 inch above the center point. Buttonholes shall be well worked over gimp to fit a 22-line button with the ends of the buttonholes securely tacked. The purling of the buttonhole shall be on the outside of the flaps. | Button hole | | 52-54 per button hole including bartack | 0 | 0 |
| | e. The finished hip pocket flap shall measure 5 1/4 ± 1/8 inches across top. | | | | | |
| 8 | Make left fly. | | | | | |
| | Finished appearance. The left fly shall finish smooth and flat without twists, puckering or distortion. The overedge stitching shall be securely caught in the fabric. | | | | | |
| | a. Position slide fastener on left fly and left fly interlining with bottom end of chain 1/4 to 1/2 inch above the fly notch. The front edge of chain shall be 1/2 to 5/8 inch from front edge of finished fly at top, tapering from 1/4 to 3/8 inch at the bottom. Stitch slide fastener tape to the fly through the interlining with a double row of stitches 3/16 to 1/4 inch gage. | 301 or 401 | LSbj-2<br>LSbj-2 | 10-14<br>10-14 | 50/3<br>50/3 | 50/3<br>50/3 |



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TABLE I MANUFACTURING REQUIREMENTS

| NO | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH/ IN | NDL | BOB/ LPR |
|---|---|---|---|---|---|---|
| 8 | Make left fly. (cont'd) | | | | | |
| | b. Overedge stitch the back edge of left fly and left fly interlining together. | 503 | SSa-1 | 6-10 | 70/2 | 70/2 |
| | | or 504 | SSa-1 | 6-10 | 70/2 | 70/2 |
| 9 | Make right fly. | | | | | |
| | Finished appearance. The right fly shall finish without bulkiness, twists, puckers or pleats. | | | | | |
| | a. The fly shall consist of one ply of basic material and lining. | | | | | |
| | b. Stitch right fly and right fly lining together 3/16 to 1/4 inch from front edge. The seam shall extend from the bottom end of fly to the upper edge of fly. | 301 | SSe-2 (a) | 10-14 | 50/3 | 50/3 |
| | | or 401 | SSe-2 (a) | 10-14 | 50/3 | 50/3 |
| | c. In the finished trousers, the upper end of the lining shall extend to top of waistband over the waistband lining, and the lining extension piece at the lower end shall not extend less than 1/2 inch beyond inseam. | | | | | |
| | d. Make a 5/8 inch long horizontal eyelet-end taper bar type buttonhole 3/8 inch below and horizontal with the waistband seam 1/2 inch in from edge of right fly piece. The buttonhole shall be well worked over gimp to fit a 22-line button. The purling of the buttonhole shall be on the underside of the fly. | Buttonhole | | 48 minimum stitches per buttonhole (including tack) | 0 | 0 |
| 10 | Assemble right and left flies. | | | | | |
| | Finished appearance. The flies shall be of even length where they join the crotch seam. The fastener parts shall be set so that there is no twisting or puckering when fly is closed. | | | | | |



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TABLE I MANUFACTURING REQUIREMENTS

| NO | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH/ IN | NDL | BOB/ LPR |
|---|---|---|---|---|---|---|
| 10 | Assemble right and left flies. (cont'd) | | | | | |
| | a. Position flies with notches at lower end matching. Seam slide fastener tape to back edge of right fly, stitching 1/4 inch from edge. When 301 stitch is used, the edge of the right fly shall be serged. | 301 or 503 or 504 | SSa-1 | 10-14 | 50/3 | 50/3 |
| 11 | Make belt loops. (see oper. 1e). | | | | | |
| | Finished appearance. The belt loops shall be flat without distortions or raw edges and with stitching of uniform gage. The loops shall be of uniform length and width. | | | | | |
| | a. Fold the stripping lengthwise towards the center, abutting the raw edges. Join the abutted edges with combination covering stitch on the underside and a row of stitches uniformly 1/16 inch from folded edges of the stripping. The stripping shall finish 5/16 to 3/8 inch wide. | 406 | EFh-1 | 10-14 | 0 | 0 |
| | -or- | | | | | |
| | b. Fold the stripping lengthwise turning the raw edges and abutting the turned under edges at the center. Join the abutted edges with a covering stitch. | 105 | EFae-1 | 10-14 | 0 | 0 |
| | -or- | | | | | |
| | c. Turn under one side of the stripping; fold the stripping lengthwise, overlapping the turned under edge on the raw edge at the center. Join the lapped edges with a covering stitch. | 105 | | 10-14 | 0 | 0 |



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TABLE I MANUFACTURING REQUIREMENTS

| NO | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH/ IN | NDL | BOB/ LPR |
|---|---|---|---|---|---|---|
| 11 | Make belt loops. (cont'd) | | | | | |
| | d. Cut seven belt loops for trousers up to size 37 and eight loops for sizes 38 and above (see operation 34). | | | | | |
| 12 | Make watch pocket (style A only). | | | | | |
| | Finished appearance. The watch pocket bearer and watch pocket halves shall be properly positioned and securely sewn without pleats, puckers, or other distortions. | | | | | |
| | a. The watch pocket shall be made of two pieces of pocketing material and a bearer of one ply of basic material. | | | | | |
| | b. Position the bearer over one ply of pocket material with the upper edges even. Turn under the lower edge of the bearer and seam to the pocket, 1/8 inch from the folded edge. A serged edge does not have to be turned. | 301 | LSd-1 | 12-16 | A | 40 |
| | c. Properly position front and back pieces of the pocket with edges even and join together at the side and bottom edges with a 1/4 inch seam. The stitching shall extend from notch to notch at upper side edges. | 301 | SSe-2 (a) | 12-16 | 40/3 | 40/3 |
| | d. Turn the pocket forcing out the seamed edges. Topstitch the pocket uniformly 1/4 inch from seamed edges. | 301 | SSe-2 (b) | 12-16 | 40/3 | 40/3 |



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| NO | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH/ IN | NDL | BOB/ LPR |
|---|---|---|---|---|---|---|
| 13 | Make and join waistband lining. | | | | | |
| | Finished appearance. The waistband lining shall measure not less than 2 inches wide in the finished trousers. The waistband linings may be pieced provided the seam finishes flat and there is no more than one piecing on each pair of trousers. The waistband stabilizer may be 1 inch shorter than the waistband lining at seat seam. In the finished trousers, the waistband lining shall not be exposed beyond the top edge of the waistband. | | | | | |
| | a. The waistband lining-interlining assembly shall consist of any one of the lining materials specified in 3.3.2.1 and one ply of interlining cotton buckram cloth specified in 3.3.3. | | | | | |
| | b. Overlap the lining on the buckram interlining 1 inch (see operation 1.j) and stitch 1/8 to 3/16 inch from edge of lining. | 301 | LSa-1 | 6-10 | 50/3 | 50/3 |
| | | or 401 | LSa-1 | 6-10 | 50/3 | 50/3 |
| | | or 101 | LSa-1 | 6-10 | 50/3 | 50/3 |
| | c. Position stabilizer on interlining with one edge of stabilizer and free edge of interlining even. Stitch stabilizer to interlining 1/2 inch from edge. | 301 | | 6-10 | 50/3 | 50/3 |
| | d. A commercial waistband lining/ interlining conforming to specified requirements may be used (see operation 1.1). | | | | | |
| 14 | Seam crotch lining to fronts. (see operation 4.d). | | | | | |
| | a. Fold crotch lining lengthwise to form a double thickness and seam to fronts or attach to fronts in overedge operation. | 301 | SSa-1 | 6-10 | 50/3 | 50/3 |
| | | or 503 | EFd-1 | 6-10 | 70/2 | 70/2 |
| | | or 504 | EFd-1 | 6-10 | 70/2 | 70/2 |



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| NO | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH/ IN | NDL | BOB/ LPR |
|---|---|---|---|---|---|---|
| 15 | Seam darts in back. | | | | | |
| | Finished appearance. The darts shall be properly positioned and finished without pleats, puckers, gathers or other distortions. In the finished trousers, the darts shall be well tapered to a flat point and the tapered end shall finish within the hip pocket at the upper edge. | | | | | |
| | a. Fold the backs, matching the cut edges of the waist darts. Stitch lengthwise along the entire length of the darts, starting 3/8 inch from the cut edges at the top end and tapering to a point at the lower end as indicated by marks on the pattern. | 301 | SSa-1 | 10-14 | 50/3 | 50/3 |
| | b. Spread open and press darts smooth and flat with a heated pressing iron or pressing machine. | | | | | |
| 16 | Attach hip pocket, flaps and label. | | | | | |
| | Finished appearance. The pockets shall finish smooth and flat without bulkiness, bulges or other distortions on the outside of the finished trousers. The stitching shall be uniform and secure. The pocket openings and flaps shall be in proper alignment. | | | | | |
| | a. Position pocket flaps on the outside of the back in line with the mark as indicated by pattern, with the pocketing on the under side. Stitch through all plies 3/16 to 1/4 inch from the mark. | 301 | LSg-2 (a) | 10-14 | 50/3 | 50/3 |
| | b. Position the hip pocket cord on the outside of trousers in line with the mark and stitch through all plies 3/16 to 1/4 inch from the mark. | 301 | SSe-2 (a) | 10-14 | 50/3 | 50/3 |



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| NO | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH/ IN | NDL | BOB/ LPR |
|---|---|---|---|---|---|---|
| 16 | Attach hip pocket, flaps and label. (cont'd) | | | | | |
| | c. Cut pocket opening through the back and pocket, simultaneously. The opening shall not extend beyond the ends of the flap. Turn the cord piece through the opening and raise stitch, forming a 1/8 to 3/16 inch cord edge at lower edge pocket opening. | 301 | SSe-1 (shows finished seam) | 10-14 | 0 or 50/3 | 0 or 50/3 |
| | d. Turn under the lower raw edge of cord facing and seam pocketing 1/16 to 1/8 inch from folded edge. When cord facing is cut on selvage, the edge need not be turned under. | 301<br>301 | LSd-1 or<br>LSbj-1 (selvage) | 10-14<br>10-14 | 50/3<br>50/3 | 50/3<br>50/3 |
| | e. Turn the pocketing wrong side out, matching notches with pocket opening, and stitch sides 3/16 to 1/4 inch from raw edges. The stitching may extend across the bottom folded edge. | 301 or 401<br>401 | SSea-2 (a)<br>SSea-2 (a) | 10-14<br>10-14 | 50-3<br>50/3 | 50/3<br>50/3 |
| | f. Turn pocketing, forcing out corners and seamed edges. Fold and stitch the pocket 1/4 to 3/8 inch from the side edges. Continue to stitch along the bottom edges. The stitching may extend across the bottom folded edge. | 301 or 301<br>301 | SSea-2 (b)<br>OSf-1 | 10-14<br>10-14 | 50/3<br>50/3 | 50/3<br>50/3 |
| | g. Turn flap down and raise stitch through all plies, 1/16 inch from joining seam, closing top of pocket. | 301 | LSq-2 (b) (show finished seams) | 10-14 | 0 or 50/3 | 0 or 50/3 |
| | h.Position the combination identification-size-instruction label on the top of the right pocketing so that on the finished trousers it will face the wearer. No part of the label printing shall be covered by the waistband interlining curtain. Stitch the label on all four sides 1/16 to 3/32 inch from edges. Stitching shall not be through printing. | 301 | LSbj-1 | 10-14 | 50/3 | 50/3 |



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| NO | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH/ IN | NDL | BOB/ LPR |
|---|---|---|---|---|---|---|
| 16 | Attach hip pocket, flaps and label. (cont'd) | | | | | |
| | -or- | | | | | |
| | i. Stitch top edge of combination identification-size-instruction label to the lower edge of the waistband lining above the right hip pocket. The ends of stitching need not be tacked provided the stitching extends at least 1/2 inch beyond each side of label. | 301 | LSa-1 | 10-14 | 0 or 50/3 | 0 or 50/3 |
| | j. Securely bartack both ends of the pocket opening through all plies of material. | 301 | SSa-1 (each end) | 10-14 | 50/3 | 50/3 |
| | -or- | | | | | |
| | k. Operations 5c, 16a, 16b and 16c may be accomplished on an automatic machine which sets the flap, stitches, cuts and turns cord facing in one operation. When this machine is used, the bearer may be trimmed to permit insertion into machine and the recommended pocket stay material may be used. | 101 | | 10-14 | 50/3 or A/3 | |
| | -and- | | | | | |
| | l. Raise stitch the joining seam of the lower edge of pocket opening through back of trousers and cord facing, only. Raise stitch the back at upper edge of pocket opening through all plies, closing the top of the pocket. Securely tack both ends of pocket opening. | 301 | | 10-14 | 0 or 50/3 | 0 or 50/3 |
| | m. The depth of the pockets shall be 5-3/4 to 6-1/4 inches when measured on the inside of pocket from top edge of cord facing to bottom. | | | | | |
| | n. The finished pocket opening (after bartacking) shall not exceed the hip pocket flap. | | | | | |



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| NO | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH/ IN | NDL | BOB/ LPR |
|---|---|---|---|---|---|---|
| 16 | Attach hip pocket, flaps and label. (cont'd) | | | | | |
| | o. On the finished trousers, the pocket flaps shall be 2-1/4 ± 1/4 inches at center measured from seam to center point. | | | | | |
| 17 | Attach watch pocket. | | | | | |
| | Finished appearance. The completed watch pocket opening shall not be stretched and the pocket material shall not be exposed on the outside. | | | | | |
| | a. Position the pocket on the outside of the right front at the notches for the pocket opening. Seam the front part of the pocket, only, to the front of trousers 3/8 inch from the top edge, from notch to notch. | 301 | SSe-2 (a) | 12-16 | 40/3 | A |
| | b. Extend depth of notches for the opening to the seam. Turn the pocket to the inside of the front of trousers and edgestitch the pocket opening 1/16 to 1/8 inch below the top edge. | 301 | SSe-2 (b) | 12-16 | A | 40/3 |
| | c. Securely tack the ends of the pocket opening through front of trousers and front and back of pocket. | 301 | SSa-1 | 12-16 | A | A |
| | d. In the finished trousers the notches for the pocket opening shall not be exposed and the edge of the opening shall meet the lower edge of the waistband. The pocket opening (after bartacking) shall be 2-7/8 ± 1/8 inches and the depth of the pocket shall be 3-3/4 ± 1/4 inches. | | | | | |



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|---|---|---|---|---|---|---|
| 18 | Join fly assembly to fronts. | | | | | |
| | Finished appearance. The seam joining the fly assembly to the fronts shall be without pleats, puckers, twists or other distortions. | | | | | |
| | a. Position left fly assembly on outside of left front with edges even and notches matching. Join left fly to front with 1/4 inch seam. | 301 | SSe-2 (a) | 10-14 | 50/3 | 50/3 |
| | b. Raise stitch the joining seam on left fly assembly 1/16 inch from seam, catching all plies of the seam. | 301 | LSq-2 (b) | 10-14 | 50/3 | 50/3 |
| | c. Position right fly assembly on right front with edges even and notches matching. Join right fly to front with 1/4 inch seam. | 301 | LSq-2 (a) | 10-14 | 50/3 | 50/3 |
| | d. Press right fly seam open and flat with heated pressing iron with the fastener tape facing towards side seam. | | | | | |
| 19 | Attach side pockets. | | | | | |
| | Finished appearance. The edges of the pocket openings shall finish smooth and flat without distortion. Pocket openings shall be of uniform appearance. The bottom front corners of the pockets shall be backstitched. Care shall be taken to avoid notching the pockets on the under side beyond the width of the side seam outlet. | | | | | |
| | a. Position the outside of the fronts on the facing of the side pocket in accordance with the notches for the pocket openings as indicated by marks on the pattern. | | | | | |



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| NO | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH/ IN | NDL | BOB/ LPR |
|---|---|---|---|---|---|---|
| 19 | Attach side pockets. (cont'd) | | | | | |
| | b. Seam front, facing and pocket material together from notch to notch 3/8 inch from the edge of the front. | 301 | SSe-2 (a) | 10-14 | 0 | 0 |
| | c. Cut the depth of the notches for the pocket opening to the seam. Raise stitch the joining seam on the pocket facing and pocket 1/16 inch from seam, catching all plies of the seam. | 301 | LSq-2 (b) | 10-14 | 50/3 | 50/3 |
| | d. Turn the pocket to the inside of the front so that the seam will finish on the inside of the pocket 1/8 inch back from the finished edge of pocket opening. | | | | | |
| | e. Top stitch the front from notch to notch, through all plies of material, 1/4 inch from the finished edge of the pocket opening. | 301 | SSe-2 (b) (show finished seam) | 10-14 | 0 | 0 |
| | f. Fold pocket inside out and seam raw edges from step to folded edge of pocketing 3/16 to 1/4 inch from edge. Notch pockets and bearers at top of step and turn. | 301 or 401 | SSae-2 (a) SSae-2 (a) | 10-14 10-14 | 50/3 or 0 50/3 | 50/3 or 0 70/2 |
| | g. Tack the ends of pocket opening to the facing with 1/4 to 3/8 inch tack. The right and left pocket opening shall not be out of alignment by more than 1/2 inch. | 301 | SSa-1 (each end) | 10-14 | 50/3 or 0 | 50/3 or 0 |
| | h. In the finished trousers the top of pocket shall extend into the waistband and be caught in the bottom stitching of waistband. | | | | | |
| | i. The finished pocket opening (after bartacking) shall measure 5-3/4 to 6-1/2 inches. | | | | | |



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|---|---|---|---|---|---|---|
| 20 | Join outseams. | | | | | |
| | Finished appearance. Outseams shall be carefully joined with notches matching, and without twists, puckers or pleats. The seams shall start and finish evenly and lie smooth and flat. | | | | | |
| | a. Join the outseam with a 3/8 inch seam, matching the notches as indicated by marks on the patterns, joining the back pocket bearer to the front at the side seam and catching the side pocket stay with its bottom end turned (if not previously caught in overedge operation) (see operation 4. b). | 301<br>or<br>401 | SSa-1<br><br>SSa-1 | 10-14<br><br>10-14 | 30/3<br><br>30/3 | 30/3<br><br>30/3 |
| | b. Spread open the outseams from top edge of trousers to bottom and press smooth and flat with a heated pressing machine with the pocket stay folded and pressed flat over the open seam. | | | | | |
| 21 | Finish side pockets. | | | | | |
| | a. Turn under the back edge of side pocket and position in line with the folded edge of the stay piece and stitch pocket to stay piece 1/4 to 3/8 inch from edge along the side seam allowance. Continue stitching along the side seamed edge of pocket to folded edge of pocket and securely back-stitch end of stitching. | 301<br>and<br>301 | SSc-1<br><br>SSe-2<br>(b) | 10-14<br><br>10-14 | 50/3<br><br>50/3 | 50/3<br><br>50/3 |
| 22 | Attach waistband and set belt loops. | | | | | |
| | Finished appearance. The finished waistband shall measure 1 3/4 ± 1/8 inches wide. There shall be seven loops for sizes up to and including 37 and eight loops for sizes 38 and up. The loops shall be positioned as follows: | | | | | |



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|---|---|---|---|---|---|---|
| 22 | Attach waistband and set belt loops. (cont'd) | | | | | |
| | One loop on each front as indicated by the mark on the pattern. | | | | | |
| | One on each side, aligned with the outseam at the back. | | | | | |
| | One centered between outseam and seat seam as indicated by the mark on the patterns. | | | | | |
| | One on the seat seam (see operation 32c and d). | | | | | |
| | The extra loop for size 38 and up on the right front in accordance with the mark on the pattern. | | | | | |
| | a. The waistband shall be made of one ply of basic material. Stitch right and left waistbands to trousers with 3/8 inch seam, catching the ends of belt loops (except center back loop, see operation 34.c and d). | 301 | SSa-1 | 10-14 | 50/3 | 50/3 |
| | b. Spread open waistband seam and press smooth and flat with a heated iron or pressing machine. | | | | | |
| | Note: The waistband lining may be attached to the waistband (operation 12) prior to setting to trousers. | | | | | |
| 23 | Join lining to waistband. | | | | | |
| | a. Turn back right fly lining and stitch top end to front and top edges of waistband. | 301 | SSe-2 (a) (each edge) | 10-14 | 50/3 | 50/3 |
| | b. Turn back the end of the left waistband in line with the joining seam of the left fly to front and stitch across top edge of waistband the width of the fly. | 301 | SSa-1 | 10-14 | 50/3 | 50/3 |



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|---|---|---|---|---|---|---|
| 23 | Join lining to waistband. (cont'd) | | | | | |
| | c. Insert top edge of waistband between lining and buckram waistband interlining with edge of lining turned in, and join all plies with a row of imitation hand-felled stitches. | 304<br>or<br>404<br>or<br>Hand | | 10-14<br><br>10-14<br><br>10-14 | 50/3<br><br>50/3<br><br>50/3 | 50/3<br><br>70/2 |
| | d. Left end of waistband lining shall finish at front edge of trousers, with the end of the waistband lining trimmed slightly back from front edge on an angle. Right end of waistband lining shall extend approximately 1 inch under the fly lining in the finished trousers. | | | | | |
| 24 | Press waistband. | | | | | |
| | a. Turn out front end of left and right waistband, forcing out the corners. Turn waistband lining to inside and press along the top edge of waistband. Trousers waistband lining shall not be exposed beyond top edge of waistband. | | | | | |
| 25 | Attach hooks and eyes. | | | | | |
| | Finished appearance. In the finished trousers, when hook and eye is engaged, the closure shall not bulge, pucker or twist. | | | | | |
| | a. Position the hook at the center (1/4 inch off center tolerance) of the left waistband with the rolled edge of hook aligned with the slide fastener scoops. Attach hook to left waistband through a suitable reinforcement. | | | | | |
| | b. Position the eye at the center of the right waistband aligned with slide fastener scoops and to correspond with hook and left waistband. Attach eye to right waistband through a suitable reinforcement. | | | | | |



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|---|---|---|---|---|---|---|
| 26 | Stitch pockets to lower edge of waistband. | | | | | |
| | a. Stitch top end of all pockets to waistband seam allowance with pockets properly positioned and and smoothed out. | 301 or 101 | SSa-1 | 10-14 | 50/3 | 50/3 |
| 27 | Finish right and left fly. | | | | | |
| | Finished appearance. The top corners of the right and left flies shall be completely forced out and shall not be bulky. The stitching of the left fly shall be uniformly stitched from edge of fly. The seam joining the left fly to the front shall not be exposed on the outside. The flies shall be smooth and flat without twists, puckers or excessive fullness. | | | | | |
| | a. Stitch down left fly with waistband lining smoothly inserted between fly and trouser. The stitching shall be 1-3/8 ± 1/8 inches from front edge of fly, extend from top of waistband, and curve to fly notch. | 301 | SSa-1 | 12-16 | 50/3 | 50/3 |
| | b. Continue the stitching of the right fly lining to the top of the waistband. | 301 | SSe-2 (a) | 12-16 | 50/3 | 50/3 |
| | | or 401 | SSe-2 (a) | 12-16 | 50/3 | 50/3 |
| | c. Turn right fly lining to the inside and stitch 1/16 to 1/8 inch from front edge. Continue the stitching along the front end of waistband, across the top of waistband and down to meet back of right fly. | 301 | SSc-2 (b) (show finished seam) | 12-16 | 0 | 0 |
| | d. Turn under the raw edge of fly lining and raise stitch the fly joining seam, catching the turned under portion of lining and slide fastener tape in the stitching. | 301 | LSq-2 (b) | 12-16 | 0 | 0 |



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|---|---|---|---|---|---|---|
| 28 | Join inseam. | | | | | |
| | Finished appearance. Inseams shall be carefully joined with notches matching, and without twists or puckers. The seams shall start and finish evenly and be smooth and flat. | | | | | |
| | a. Join the inseam from the bottom edge of the legs to the crotch with the notches matching as indicated by marks on the pattern, forming an outlet, tapering the seam allowance of the back from 7/8 inch at the crotch to 3/8 inch at the knee notch. | 301 | SSbd-1 (above knee notch) | 12-16 | 30/3 | 30/3 |
| | | 301 or | SSa-1 (below knee notch) | 12-16 | 30/3 | 30/3 |
| | | 401 | SSbd-1 (above knee notch) | 12-16 | 30/3 | 50/3 |
| | | 401 | SSa-1 (below knee notch) | 12-16 | 30/3 | 5 |
| | b. Spread open inseam and press smooth and flat with a heated pressing iron or pressing machine. | | | | | |
| 29 | Join crotch and seat seams. | | | | | |
| | Finished appearance. The crotch and seat seams shall be joined with stitch tension of sufficient elasticity to meet strain encountered in sitting and stooping. The waist shall be measured before joining crotch and seat seam to assure correct waist size. In the finished trousers, the inseams shall meet and right and left waistband joining seams shall be aligned; the base of the left fly shall cover the right fly on the outside. | | | | | |
| | a. Measure waist to proper size before joining crotch, seat and waist seam. | | | | | |



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|---|---|---|---|---|---|---|
| 29 | Join crotch and seat seams. (cont'd) | | | | | |
| | b. Join crotch and seat seams from the base of the fly through the curve in the crotch with a 1/4 inch seam, then taper to the end of waistband lining with a double needle seat seamer with the rows of stitching 1/64 to 1/32 inch apart. The seams from the waistband joining seam to the end of the waistband lining shall be perpendicular (90 degree angle) to the waistband joining seam. | 401 or 401 | SSa-2 each needle SSa-2 | 8-10 8-10 each needle | 30/3 AA Polyester | 50/3 A Polyester |
| | c. Spread open crotch and seat seams and press smooth and flat with a heated pressing iron or machine. | | | | | |
| | d. The inseam at the crotch shall not stagger more than 3/8 inch. | | | | | |
| 30 | Finish waistband. | | | | | |
| | a. Top stitch the spread open waistband seam 1/16 inch below the joining seam, catching the waistband lining (with ends of center back folded in) and the top of pockets in the stitching. The stitching shall not extend across the fly. | 301 | LSbd-2 (show finished seam) | 12-16 | 50/3 | 50/3 |
| 31 | Assemble fasteners. | | | | | |
| | a. Attach slider and with ends of slide fastener chain even, securely staple bottom end of chain together. | | | | | |



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|---|---|---|---|---|---|---|
| 32 | Press seams (with heated pressing iron or pressing machine). | | | | | |
| | Finished appearance. The seams shall finish smooth and flat without pleats, puckers or twists. | | | | | |
| | a. Spread open the seat and crotch seams from top of waistband to fly notch and press smooth and flat. | | | | | |
| 33 | Stitch right fly extension. | | | | | |
| | Finished appearance. In the finished trousers the right fly lining extension shall finish not less than 1/2 inch beyond juncture point of the right and left inseam. | | | | | |
| | a. Turn the raw edges and end of the right fly lining extension and machine stitch or hand fell to to each side of the open crotch seat seam, catching the turned under end in the stitching. | 301 or Hand | SSac-3 (b) (c) and (d) SSac-3 (b) (c) and (d) | 12-16 6-8 | 50/3 50/3 | 50/3 |
| 34 | Finish belt loop attachments. | | | | | |
| | Finished appearance. All belt loop openings shall measure 2 ± 1/8 inches from bartacking of loop to folded edge. The tacking shall extend the width of the loops, not beyond. All loops shall be vertically straight. The center back loop shall be centered on the waistband and seat seams. | | | | | |
| | a. Position belt loops flat against the trousers with underside of loops facing up. Securely stitch or bartack loops to trousers 1/4 to 5/16 inch below waistband trouser seam. The tacking shall not be visible in the finished trousers. | 301 or Bartack or Automatic loop tacker | SSa-1 | 12-16 21-28 (per bartack) 24-28 (per bartack) | 0 0 0 | 0 0 |



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TABLE I MANUFACTURING REQUIREMENTS

| NO | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH/ IN | NDL | BOB/ LPR |
|---|---|---|---|---|---|---|
| 34 | Finish belt loop attachments. (cont'd) | | | | | |
| | b. Turn under the top end of loops and align folded top end of loops with top edge of waistband. Securely stitch or bartack the belt loops 1/8 inch below the top folded edge of loops at top of waistband with the turned under portion caught in the stitching. The stitching or bartack shall extend the width of the loops. | 301 or Bartack or Auto-matic loop tacker | LSd-1 | 12-16<br>21-28 (per bartack)<br>24-28 (per bartack) | 0<br>0<br>0 | 0<br>0<br>0 |
| | c. Position back loop over the seat seam, with the right side of loop against the trousers, and with one end of loop positioned 5/8 to 3/4 inch from the top end of the waistband. Stitch or bartack across the loop 1/8 to 3/16 inch from raw end of loop. Turn up the loop, covering the top raw edge, and stitch or bartack the loop above the first tacking. | 301 or Bartack | LSq-2 (a) | 12-16<br>21-28 (per bartack) | 0<br>0 | 0<br>0 |
| | d. Fold loop back even with top edge of waistband. Turn under bottom raw end and stitch or bartack to trousers 1/4 to 5/16 inch below waistband seam. | 301 or Bartack | LSd-1 | 12-16<br>21-28 (per bartack) | 0<br>0 | 0<br>0 |
| 35 | Bartack trousers. | | | | | |
| | Finished appearance. The trousers shall not be distorted by the bartacks. The pockets shall lie smooth and flat. The bartacks shall be properly positioned and correctly placed to reinforce the points of strain as specified. | | | | | |
| | a. Bartack both ends of all pocket openings. Bartacks for the back pockets shall be vertical. Bartacks for the side pockets shall be horizontal. | Bartack | | 21-28 (per bartack) | 0 | 0 |



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TABLE I MANUFACTURING REQUIREMENTS

| NO | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH/ IN | NDL | BOB/ LPR |
|---|---|---|---|---|---|---|
| 35 | Bartack trousers. (cont'd) | | | | | |
| | b. Bartack the bottom end of the fly horizontally through all plies above the juncture of the crotch seam and fly. The bartack shall not extend beyond crotch seam. | Bartack | | 21-28 (per bartack) | 0 | 0 |
| | c. Bartack the inside front edge of right fly to back of left fly with a vertical bartack, with lower end of bartack approximately 1/2 inch above the horizontal bartack at base of fly. | Bartack | | 21-28 (per bartack) | 0 | 0 |
| | d. Bartacks shall be made 3/8 to 1/2 inch long, and shall be stitched through all plies of material at the point of bartacking. | | | | | |
| 36 | Mark, or gage, and sew buttons. | | | | | |
| | Finished appearance. The buttons shall be securely sewn and properly aligned with buttonholes. Hand sewn buttons shall be shank wrapped three times and the ends of the thread shall be securely tacked with not less than two stitches. Machine sewn buttons which are automatically shanked shall have a minimum of ten wraps with three locking stitches. Shank wrapping is optional on machine sewn buttons. | | | | | |
| | a. Sew a 22-line button on each hip pocket through top ply of pocketing to correspond with buttonhole in pocket flap. | 301 or 101 or Hand | | 14-16 per button 14-16 per button 4 double per button not including shank wrapping | 20/4 20/4 16/4 | 20/4 0 |



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TABLE I MANUFACTURING REQUIREMENTS

| NO | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH/ IN | NDL | BOB/ LPR |
|---|---|---|---|---|---|---|
| 36 | Mark, or gage, and sew buttons. (cont'd) | | | | | |
| | b. Sew a 22-line button on the left waistband lining to correspond with the right fly buttonhole. The stitching shall not be visible on the outside of the trousers. | 301 or | | 14-16 per button | 20/4 | 20/4 |
| | | 101 or | | 14-16 per button | 20/4 | |
| | | Hand | | 4 double per button | 16/4 | |
| | | | | not including shank wrapping 7 (minimum) 1 deep set per button | 50/3 | |
| 37 | Press trousers. | | | | | |
| | Finished appearance. All pressing shall be done with a heated pressing iron or pressing machine of a type suitable to obtain the desired appearance in accordance with good manufacturing practice. | | | | | |
| | a. Press the top, flies and pockets smooth and flat. | | | | | |
| | b. Press smooth and crease the legs in the center with the seams matching at the bottom and in the crotch or the inseam in the crotch may be slightly back of the outseam. | | | | | |
| | c. Close flies, waistband and pockets. | | | | | |
| 38 | Finish trouser bottom. | | | | | |
| | a. Overedge stitch or pink all trouser bottoms. | | | | | |



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TABLE I MANUFACTURING REQUIREMENTS

| NO | OPERATION | STCH TYPE | SEAM/ STCH TYPE | STCH/ IN | NDL | BOB/ LPR |
|---|---|---|---|---|---|---|
| 39 | Clean trousers.<br><br>a. Trim all ends of stitching and remove all loose threads, spots, stains and shade tickets or markings without injury to the material.<br><br>b. Trim ends of belt loops to a length of not more than 3/8 inch. | | | | | |

3.7 Measurements. The finished trousers shall conform to the measurements shown in Table II.

TABLE II - MEASUREMENTS IN INCHES 1/

| Size | Waist 2/ | Seat 3/ | Knee 4/ | | Bottom 5/ | |
|---|---|---|---|---|---|---|
| | | | Short X-Short | Regular, Long, X-Long | Short X-Short | Regular Long, X-Long |
| 26 | 26 | 35 | 18-1/2 | 19 | 17-1/2 | 18 |
| 28 | 28 | 37 | 19 | 19-1/2 | 18 | 18-1/2 |
| 30 | 30 | 39 | 19-1/2 | 20 | 18-1/2 | 19 |
| 32 | 32 | 41 | 20 | 20-1/2 | 19 | 19-1/2 |
| 34 | 34 | 43 | 20-1/2 | 21 | 19-1/2 | 20 |
| 36 | 36 | 45 | 20-1/2 | 21 | 19-1/2 | 20 |
| 38 | 38 | 47 | 21 | 21-1/2 | 20 | 20-1/2 |
| 40 | 40 | 49 | 21 | 21-1/2 | 20 | 20-1/2 |
| 42 | 42 | 51 | 21-1/2 | 22 | 20-1/2 | 21 |
| 44 | 44 | 53 | 21-1/2 | 22 | 20-1/2 | 21 |
| 46 | 46 | 55 | 22 | 22-1/2 | 21 | 21-1/2 |
| Tolerance | +3/4 -1/2 | +3/4 -1/2 | ± 1/2 | ± 1/2 | ± 1/2 | ± 1/2 |

Inseams: 6/

| | |
|---|---|
| X-Short | 32 inches ± 1/2 |
| Short | 34 inches ± 1/2 |
| Regular | 36 inches ± 1/2 |
| Long | 38 inches ± 1/2 |
| X-Long | 40 inches ± 1/2 |



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3.7.1 Method of measuring. The trousers shall be fastened and placed flat upon a table except as otherwise indicated herein.

1/ Measurements in inches.

2/ Twice the measurement taken across top edge of waistband from folded edge to folded edge.

3/ Twice the measurement across trousers, 1 inch above fly bartack, from folded edge to folded edge.

4/ The location of knee measurements shall be 12-1/2 inches below crotch seam for X-Short, 13-1/2 inches for Shorts, 14-1/2 inches for Regulars, 15-1/2 inches for Longs and 16-1/2 inches for X-Longs.

5/ Measure along (unhemmed) bottom edge of leg from folded edge to folded edge.

6/ Measure along inseam from crotch seam to unhemmed bottom edge. (An allowance of 3 inches for hems is included in the inseam length.)

3.8 Workmanship. The finished trousers shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirement.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4).
2. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in MIL-STD-1488.

4.4 First article inspection. When first article is required, inspection shall be made of completely fabricated trousers for all provisions of this specification.

4.5 Quality conformance inspection. Inspection shall be performed in accordance with MIL-STD-1488.



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4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document. The contractor shall furnish a certificate of compliance stating that the hooks and eyes are one of the specified types, or, if an "equal" type is used, that it conforms in all respects to that approved by the contracting officer as required in 3.3.6. In addition, the requirement that all components of the slide fastener have been furnished by one supplier shall be accepted on a certificate of compliance (see 3.3.5).

4.5.2 End item examination. Samplings and examination of the end item shall be performed for dress (D) trousers in accordance with the provisions of MIL-STD-1488.

4.5.3 Dimensional examination. The dimensional examination provisions for the finished trousers shall be as listed in 3.7 and as specified in MIL-STD-1488.

5. PACKAGING.

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Each pair of trousers shall be folded so that the length dimension is approximately 23 inches.

5.1.2 Commercial. Trousers shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.2.1 Level A. Thirty-five pair of trousers of one size and style only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container, assembled, closed, and reinforced conforming to class weather-resistant, variety DW, grade V15c of MIL-B-17757. Level A folded trousers shall be packed flat, alternated end for end, within a shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound (13.6 kg) minimum basis weight kraft paper conforming to type I, grade B of UU-P-268.

5.2.2 Commercial. Trousers, preserved as specified in 5.1, shall be packaged in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or purchase order, interior packages (when applicable) and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The trousers covered by this specification are intended for wear by male personnel of the U. S. Marine Corps.



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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Style and size required (see 1.2).
(c) Whether first article is required (see 3.2).
(d) Number of trousers in first article (see 3.2).

6.3 Samples and shade samples. For information regarding the availability of sample of the trousers and of standard shades specified, address inquiry to the acquiring activity issuing the Invitation for Bids.

6.4 First article. First article examination, test and approval shall be as specified by the contracting officer (see 3.2).

6.5 Figures. Figure 1 is furnished for information only. To the extent of any inconsistencies between the written specification and the figure, the specification shall govern.

6.6 Subject term (keyword) listing.
Men's
Polyester
Trousers
Wool

Preparing Activity:
Navy - MC
Project No. 8405-N243



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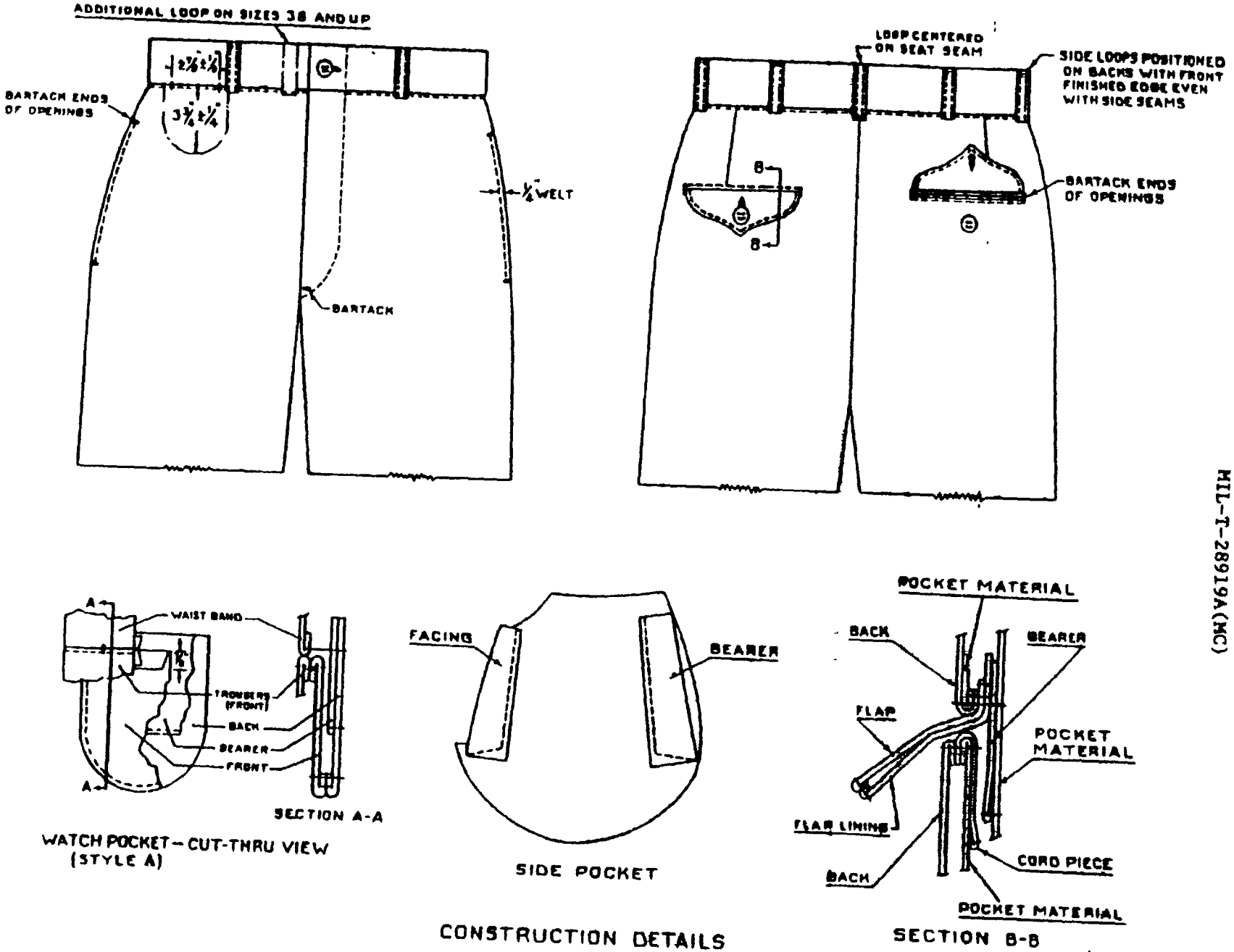



FIGURE 1. TROUSERS, MEN'S: SERVICE, SUMMER AND WINTER (OFFICER'S)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER: MIL-T-28919A(MC) | 2. DOCUMENT DATE (YYMMDD): 20 April 1994 |
|---|---|---|

3. DOCUMENT TITLE: TROUSERS, MEN'S: SERVICE, SUMMER AND WINTER (OFFICER'S)

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

| 6. SUBMITTER | | | |
|---|---|---|---|
| a. NAME (Last, First, Middle Initial) | | b. ORGANIZATION | |
| c. ADDRESS (Include Zip Code) | | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) | 7. DATE SUBMITTED (YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME | b. TELEPHONE (Include Area Code)<br>(1) Commercial | (2) AUTOVON |
|---|---|---|
| COMMANDER | (912) 436-6560 | 567-6560 |
| c. ADDRESS (Include Zip Code)<br>MARINE CORPS LOGISTICS BASE, CODE 855<br>814 RADFORD BOULEVARD<br>ALBANY GA 31704-1128 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041 3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |

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